Batching and epochs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

TEACHER GUIDE · 1 OF 9

Prepare the purpose, prerequisites and materials

> Prepare Batching and epochs: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Distinguish an example a **batch** an **update** and an **epoch**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Training a small network

Without opening the old lesson, explain one key idea from **Training** a small network. Give an example and a mistake that the idea helps you avoid.

Look for: Trace repeated forward loss gradient and **update** steps

Reference example, not the only acceptable wording: Teacher explanation: **Prediction**s [0, 0], errors [-2, -4], MSE 10, gradient -10 and new v = 1.

Gradients with a numerical check

Without opening the old lesson, explain one key idea from Gradients with a numerical check. Give an example and a mistake that the idea helps you avoid.

Look for: Estimate one partial derivative while other parameters stay fixed

Reference example, not the only acceptable wording: Teacher explanation: **Prediction** 3, residual 0, loss 0, gradient [0, 0].

Parameters and architecture

Without opening the old lesson, explain one key idea from Parameters and architecture. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish architectural choices from parameter values

Reference example, not the only acceptable wording: Teacher explanation: There are 8 weights and 2 biases, total 10.

For the compact alternative schedule, use two 45-minute sessions to separate counting from numerical **training**. Keep the frozen hidden layer from Week 12. Have students move paper record cards into **batch**es before doing arithmetic.

The coincident final values for the proportional-target example require the independent counterexample; do not teach that ordering is generally irrelevant.

Outcomes

Distinguish an example a **batch** an **update** and an **epoch**

Trace parameter changes across successive **batch**es

Calculate **update** counts and aggregate unequal **batch** losses correctly

Preparation

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Prerequisite answer

Four **update**s occurred. State zero was the initial model before **training**.

End goal for the pictorial case

By the end, I can explain **batch**ing and **epoch**s using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain and model the mechanism

> Model Batching and epochs using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Explain Batching and epochs · 1

In Week 12 every **update** used both **training** examples. We now keep the same network h = ReLU(x), y = vh, with only v trained, and divide the data into **batch**es. In this lesson each nonempty **batch** produces exactly one **update**.

An **epoch** visits every **training** example exactly once, without replacement. These rules make the counts unambiguous.

Imagine five fictional records A through E in fixed order. With **batch size** 2, the groups are [A, B], [C, D] and [E]. Keep the smaller last **batch**.

One **epoch** has three **batch**es and three **update**s; two **epoch**s have six **update**s and ten example visits.

When the last **batch** is retained, **update**s per **epoch** equal the row count divided by **batch size** rounded up. A program that drops the final incomplete **batch** follows a different rule and omits examples in that pass.

Record the rule instead of silently changing the denominator.

Explain Batching and epochs · 2

Use the two fictional pairs A = (1, 2) and B = (2, 4), start v = 0 and rate 0.1. For one pair with hidden value h, squared-error gradient is 2 times (vh - target) times h.

For a **batch**, average these gradients at the same current v. Apply one **update**, then recompute for the next **batch**.

Explain Batching and epochs · 3

Assume finite numeric pairs, integer size and finite rate. With size 1, A gives gradient -4 and new v = 0.4. B then uses 0.4, giving **prediction** 0.8, residual -3.2 and gradient -12.8, so v becomes 1.68.

With size 2, both gradients are evaluated at v = 0; their mean is -10, producing v = 1.

Term	Meaning
Batch	A group of training examples used together to calculate one gradient here
Update	One application of the gradient and learning rate to the current parameters
Epoch	One pass through all training examples under the stated sampling rule
Batch size	The intended number of examples in each batch with a possible smaller final batch

One pass can contain different numbers of updates

BATCH RULE	PARAMETER AFTER EACH BATCH	UPDATES
A then B separately	0 to 0.4 to 1.68	2
A and B together	0 to 1	1
B then A separately	0 to 1.6 to 1.68	2

All three passes visit two examples. The reversed order happens to end at the same value for these proportional targets; that coincidence is not a general order-invariance rule.

Guided: B at v = 0 gives gradient -16 and v = 1.6. A then gives residual -0.4, gradient -0.8 and v = 1.68. Two **batch**es each cause one **update**. Exit: both teams visit ten examples.

Size 2 produces six **update**s; size 5 produces two. Record both budgets and all **training** settings.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

One pass can contain different numbers of updates

BATCH RULE	PARAMETER AFTER EACH BATCH	UPDATES
A then B separately	0 to 0.4 to 1.68	2
A and B together	0 to 1	1
B then A separately	0 to 1.6 to 1.68	2

All three passes visit two examples. The reversed order happens to end at the same value for these proportional targets; that coincidence is not a general order-invariance rule.

For B then A separately, show both gradients and new parameter values. Why does one epoch still contain two updates?

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Batching and epochs

Show why an **epoch**'s example order can matter for single-example **update**s.

The facts we will use

Supplied facts

Toy y=wx, rate 0.1, squared loss, start w = 0. **Training** A=(x = 1, target 2), B=(x = 2, target 0).

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/13

Batching and epochs

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. A then B

A gradient -4 gives w = 0.4; B gradient 3.2 then gives w = 0.08.

2. Restart and reverse

B at w=0 has gradient 0; A then gives w = 0.4.

3. Compare full batch

At w = 0 mean gradient=(-4+0)/2=-2; one full-batch update gives w = 0.2.

CHANGE ONE THING

Use only one full-batch update after reversing record order.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. A then B	A gradient -4 gives w = 0.4; B gradient 3.2 then gives w = 0.08.
2. Restart and reverse	B at w=0 has gradient 0; A then gives w = 0.4.
3. Compare full batch	At w = 0 mean gradient=(-4+0)/2=-2; one full-batch update gives w = 0.2.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A single-example **update** changes the weight before the next example's gradient is calculated. Reversing the example order can therefore change the path.

A full-**batch update** computes both gradients at the same starting weight and averages them before changing that weight. The distinction is when the **update** occurs, not whether the examples have different labels.

What this example does not establish

Keep the stated limit

Different order effects do not mean a universally better **batch**ing strategy; compare

declared schedules and budgets fairly.

Change one thing in the pictured case

Change one thing

Change: Use only one full-**batch update** after reversing record order.

Predicted result

Prediction: The **update**d weight remains 0.2.

Why the result follows

Reason: The same mean of the two starting gradients is unchanged by their order.

Ask while showing the picture

Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Batching and epochs, reduce support deliberately and check individual reasoning before discussing answers.

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve states versus updates, then arrange five cards into batches of two.
7 to 18 minutes	Count batches, updates and visits for the table and W1; discuss retaining the final group.
18 to 32 minutes	Trace A then B at current v. Complete W2 and identify exactly where the stale gradient arose.
32 to 45 minutes	Run or trace the code and compare the full-batch update with the two singleton updates.
Session 2 start to 8 minutes	Complete the guided reverse-order trace and mark its coincidental equal endpoint.
8 to 20 minutes	Calculate fixed-model loss and use W3 to weight unequal batch means.
20 to 35 minutes	Collect W4 independently, requiring a fresh reset for the reverse-order run.
35 to 45 minutes	Discuss W5 and the exit response, then assess the fresh retry after targeted feedback.

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide separate boxes for old v, **batch** gradient and new v; allow calculators. Extension: at a fixed v, prove that the row-count-weighted mean of **batch** gradients equals the full-data mean gradient.

Explain why successive **update**s with changed v are a different procedure.

Reduce help deliberately

Completion practice

Completion task: For A then B separately in the lesson, a learner computes both gradients at v = 0, then applies them to get v = 2. Explain the error and give the correct final value.

Reasoning to listen for

Teacher check: Teacher explanation: The second gradient must use v = 0.4. Using -16 from the old state is stale; the current B gradient is -12.8 and the final v is 1.68.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Batch**ing and **epoch**s. Proposed explanation: “One **epoch** always means one **update**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Only a full-data **batch** gives one **update** per **epoch** in this procedure.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why do the single-example orders finish at different weights? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

The second gradient is evaluated at an already **update**d weight. Full-**batch** gradients here share the same starting weight before averaging. Changed-case check: The **update**d weight remains 0.2.

The same mean of the two starting gradients is unchanged by their order.

Changed-case reasoning: The **update**d weight remains 0.2. The same mean of the two starting gradients is unchanged by their order.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

TEACHER GUIDE · 4 OF 9

Run the investigation and diagnose misconceptions

> Run the Batching and epochs investigation and use the first unsupported step to choose a teaching response.

One **epoch** always means one **update**: Only a full-data **batch** gives one **update** per **epoch** in this procedure.

Reuse all gradients from the start of an **epoch**: Recompute at the parameter value before each **batch update**.

Average **batch** means equally: At one fixed model, weight each **batch** mean by the number of examples in that **batch**.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows	Next teaching action
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Batching and epochs questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Guided and exit questions

Guided: B at v = 0 gives gradient -16 and v = 1.6. A then gives residual -0.4, gradient -0.8 and v = 1.68. Two **batch**es each cause one **update**. Exit: both teams visit ten examples.

Size 2 produces six **update**s; size 5 produces two. Record both budgets and all **training** settings.

W1 Count a partial batch

Workbook question · W1

For 10 rows, **batch size** 4 and three **epoch**s, retaining the final **batch**, list **batch** lengths and count total **update**s and example visits.

Teacher reasoning · W1

Teacher explanation: **Batch** lengths 4, 4, 2; three **update**s per **epoch**, nine total **update**s and 30 example visits.

W2 Repair a stale gradient

Workbook question · W2

For A then B separately in the lesson, a learner computes both gradients at v = 0, then applies them to get v = 2. Explain the error and give the correct final value.

Teacher reasoning · W2

Teacher explanation: The second gradient must use v = 0.4. Using -16 from the old state is stale; the current B gradient is -12.8 and the final v is 1.68.

W3 Combine unequal batches

Workbook question · W3

At one frozen model, **batch**es of sizes 3 and 1 have mean squared errors 2 and 10. Find the full MSE and explain why the average of 2 and 10 is wrong.

Teacher reasoning · W3

Teacher explanation: Full MSE is (3 times 2 + 10) / 4 = 4. The unweighted average 6 gives the single-row **batch** too much weight.

W4 Test order on a new case independently

Workbook question · W4

Use A = (1, 1), B = (2, 0), h = ReLU(x), y = vh, v = 0 and rate 0.1. Train with size 1 for one **epoch** in A then B order, then restart at 0 and reverse the order. Show gradients and parameter values. Compare the final values.

Teacher reasoning · W4

Teacher explanation: A then B: A gradient -2 gives v = 0.2; B residual 0.4 gives gradient 1.6 and v = 0.04. B then A: B gradient 0 keeps v = 0; A gradient -2 gives v = 0.2. Both see the same rows but finish differently.

W5 Audit an epoch claim

Workbook question · W5

A program **process**es 11 rows using size 4 and drops the last incomplete **batch**. How many examples did it visit and how many **update**s occurred? What must its report disclose?

Teacher reasoning · W5

Teacher explanation: It visits eight rows in two **batch**es and performs two **update**s. Three rows are omitted; it did not complete a full pass under this lesson definition. Report the drop rule and actual visits.

Independent assessment and fresh retry

Teacher key · Independent assessment and fresh retry

W4 requires the two traces and final values 0.04 and 0.2. Fresh retry: A = (1, 2), B = (1, 0), start v = 0, rate 0.25, singleton **batch**es. A then B gives gradients -4 and 2, values 1 and 0.5.

Reverse order gives gradients 0 and -4, values 0 and 1. Ask students to name the reset and explain the order effect.

Assess independent transfer

Independent question · withhold the key

Selected case: Use A = (1, 1), B = (2, 0), h = ReLU(x), y = vh, v = 0 and rate 0.1. Train with size 1 for one **epoch** in A then B order, then restart at 0 and reverse the order. Show gradients and parameter values.

Compare the final values.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: A then B: A gradient -2 gives v = 0.2; B residual 0.4 gives gradient 1.6 and v = 0.04. B then A: B gradient 0 keeps v = 0; A gradient -2 gives v = 0.2. Both see the same rows but finish differently.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect the project, support and next lesson

> Connect Batching and epochs to Image-pattern Network Investigation, adapt support and plan a later changed-case check.

Add a labelled **batch**ing and **epoch**s **evidence** sheet to Image-pattern Network Investigation. Show how you can distinguish an example a **batch** an **update** and an **epoch**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide separate boxes for old v, **batch** gradient and new v; allow calculators. Extension: at a fixed v, prove that the row-count-weighted mean of **batch** gradients equals the full-data mean gradient.

Explain why successive **update**s with changed v are a different procedure.

Week 14 compares learning rates under one fixed **batch** rule.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-5/14, ai-5/16, ai-5/19. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Batching and epochs lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Distinguish an example a **batch** an **update** and an **epoch**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: For A then B separately in the lesson, a learner computes both gradients at v = 0, then applies them to get v = 2. Explain the error and give the correct final value.

Supported practice answer

Reasoned practice answer: Teacher explanation: The second gradient must use v = 0.4. Using -16 from the old state is stale; the current B gradient is -12.8 and the final v is 1.68.

Check every learner before discussion: Use the worked case for **Batch**ing and **epoch**s. Proposed explanation: “One **epoch** always means one **update**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Only a full-data **batch** gives one **update** per **epoch** in this procedure.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Use only one full-**batch update** after reversing record order. Require a **prediction** and a reason before comparing. Expected result: The **update**d weight remains 0.2.

Reason: The same mean of the two starting gradients is unchanged by their order.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Batching and epochs graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Batching and epochs: the case

AI ACADEMY / AI-5/13 · 01 · LOOK AT THE FACTS

Batching and epochs: the case

The situation

Show why an epoch's example order can matter for single-example updates.

Supplied evidence

Toy y=wx, rate 0.1, squared loss, start w = 0. Training A=(x = 1, target 2), B=(x = 2, target 0).
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain batching and epochs using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

Show why an **epoch**'s example order can matter for single-example **update**s. Supplied facts: Toy y=wx, rate 0.1, squared loss, start w = 0. **Training** A=(x = 1, target 2), B=(x = 2, target 0).

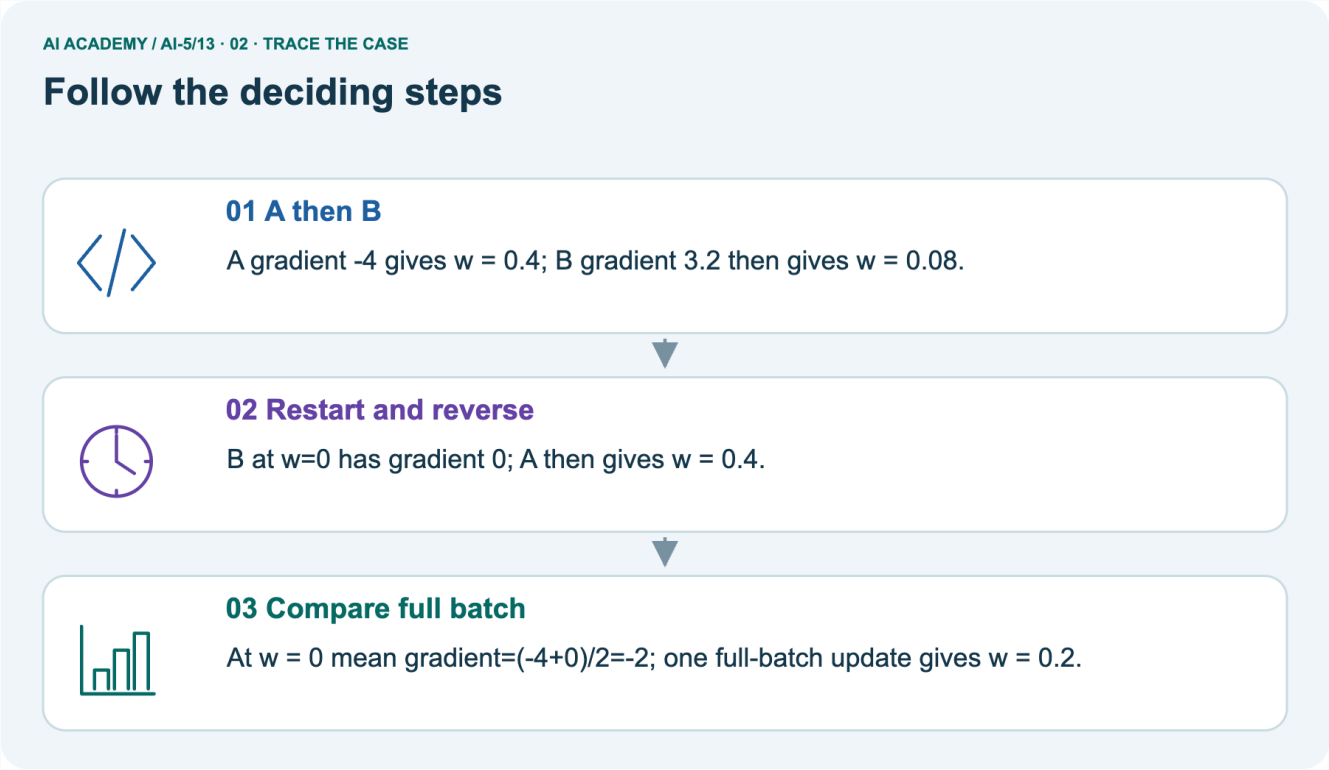
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

A then B: A gradient -4 gives w = 0.4; B gradient 3.2 then gives w = 0.08. → Restart and reverse: B at w=0 has gradient 0; A then gives w = 0.4. → Compare full **batch**: At w = 0 mean gradient= (-4+0)/2=-2; one full-**batch update** gives w = 0.2.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. One pass can contain different numbers of updates

AI ACADEMY / AI-5/13 · 03 · INSPECT THE LESSON MODEL

One pass can contain different numbers of updates

Batch rule	Parameter after each batch	Updates
A then B separately	0 to 0.4 to 1.68	2
A and B together	0 to 1	1
B then A separately	0 to 1.6 to 1.68	2

All three passes visit two examples. The reversed order happens to end at the same value for these proportional targets; that coincidence is not a general order-invariance rule.

Follow the labelled picture with the matching explanation. Open the vector diagram

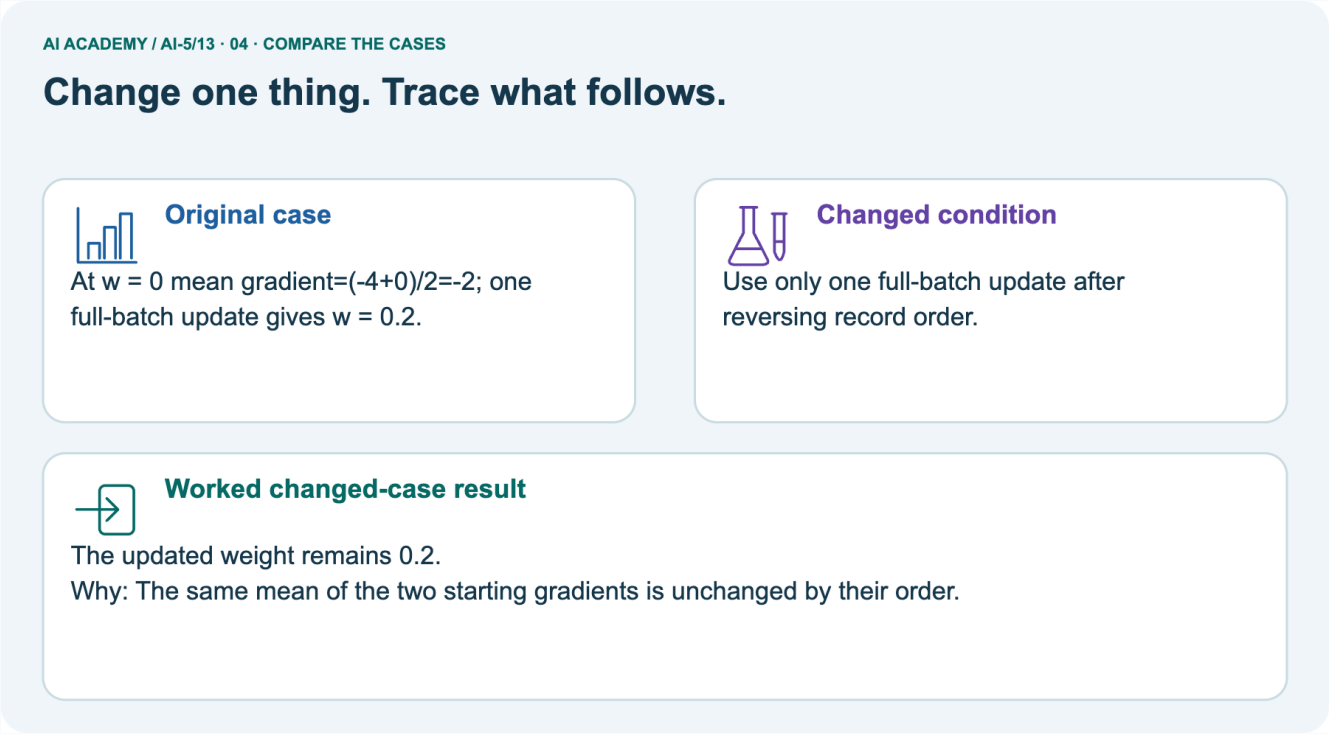
One pass can contain different numbers of **update**s. A then B separately | 0 to 0.4 to 1.68 | 2. A and B together | 0 to 1 | 1. B then A separately | 0 to 1.6 to 1.68 | 2

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: At w = 0 mean gradient=(-4+0)/2=-2; one full-**batch update** gives w = 0.2. Changed condition: Use only one full-**batch update** after reversing record order. Expected result: The **update**d weight remains 0.2.

Reason: The same mean of the two starting gradients is unchanged by their order.

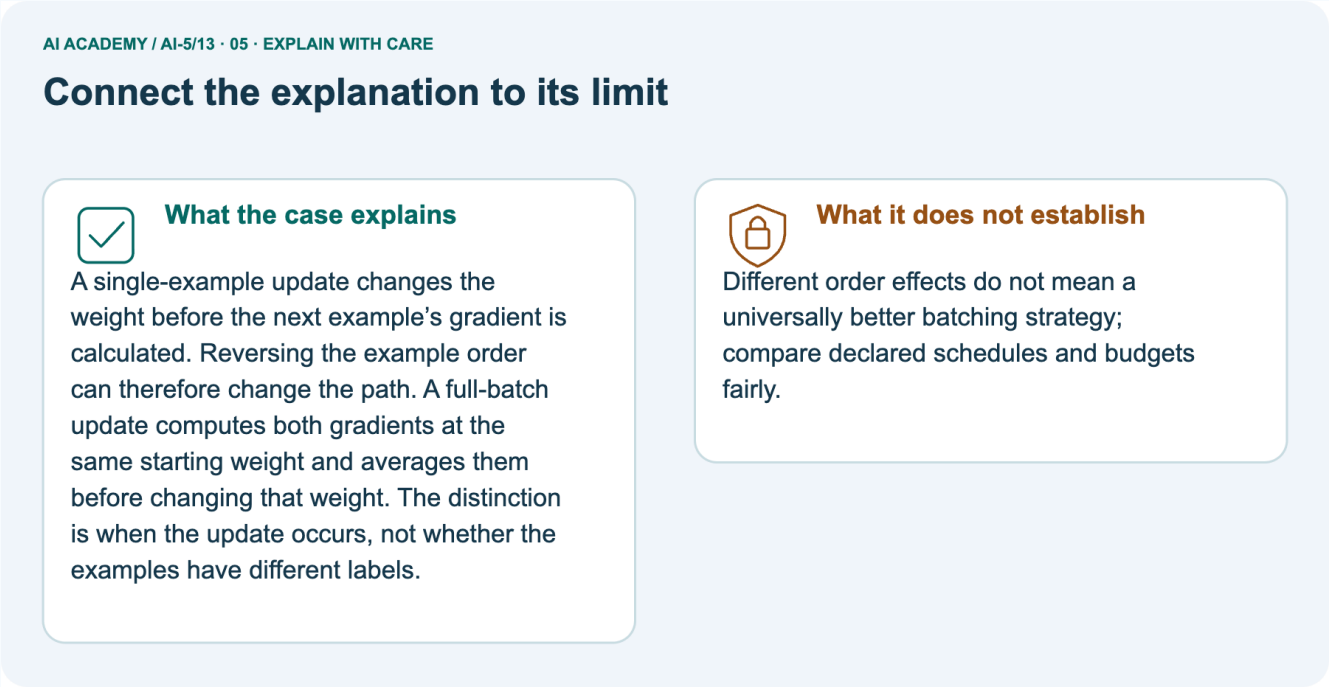
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The same mean of the two starting gradients is unchanged by their order.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A single-example **update** changes the weight before the next example's gradient is calculated. Reversing the example order can therefore change the path.

A full-**batch update** computes both gradients at the same starting weight and averages them before changing that weight. The distinction is when the **update** occurs, not whether the examples have different labels.

Limit: Different order effects do not mean a universally better **batch**ing strategy; compare declared schedules and budgets fairly.

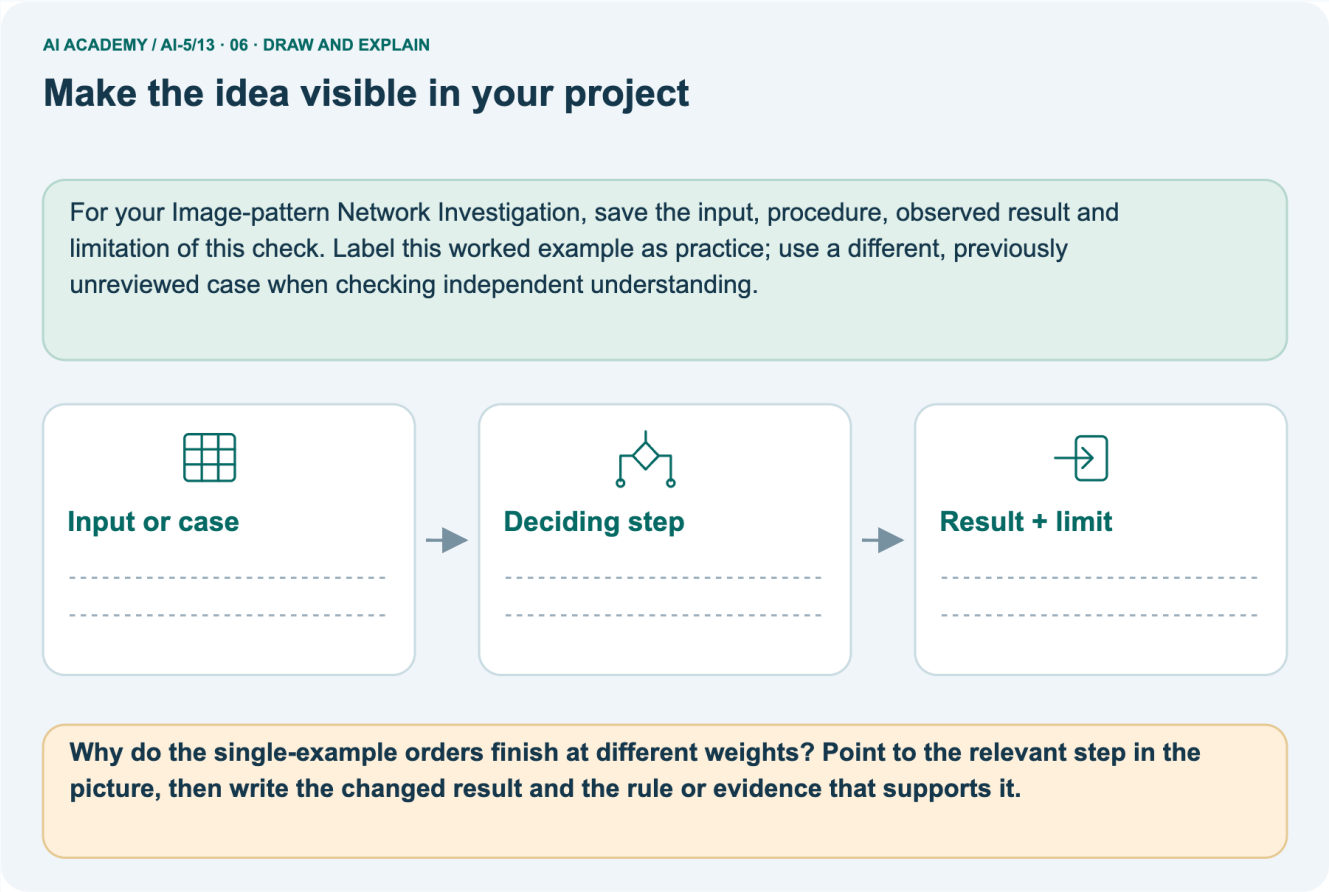
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Different order effects do not mean a universally better **batch**ing strategy; compare declared schedules and budgets fairly.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why do the single-example orders finish at different weights? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The second gradient is evaluated at an already **update**d weight. Full-**batch** gradients here share the same starting weight before averaging. Changed-case check: The **update**d weight remains 0.2.

The same mean of the two starting gradients is unchanged by their order.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.